

SOFTWARE ANALYSIS DOCUMENTATION SEQUENCE DIAGRAMS

RENTAMOVIL

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ANÁLISIS Y DESARROLLO DE SOFTWARE 3145556

SERVICIO NACIONAL DE APRENDIZAJE

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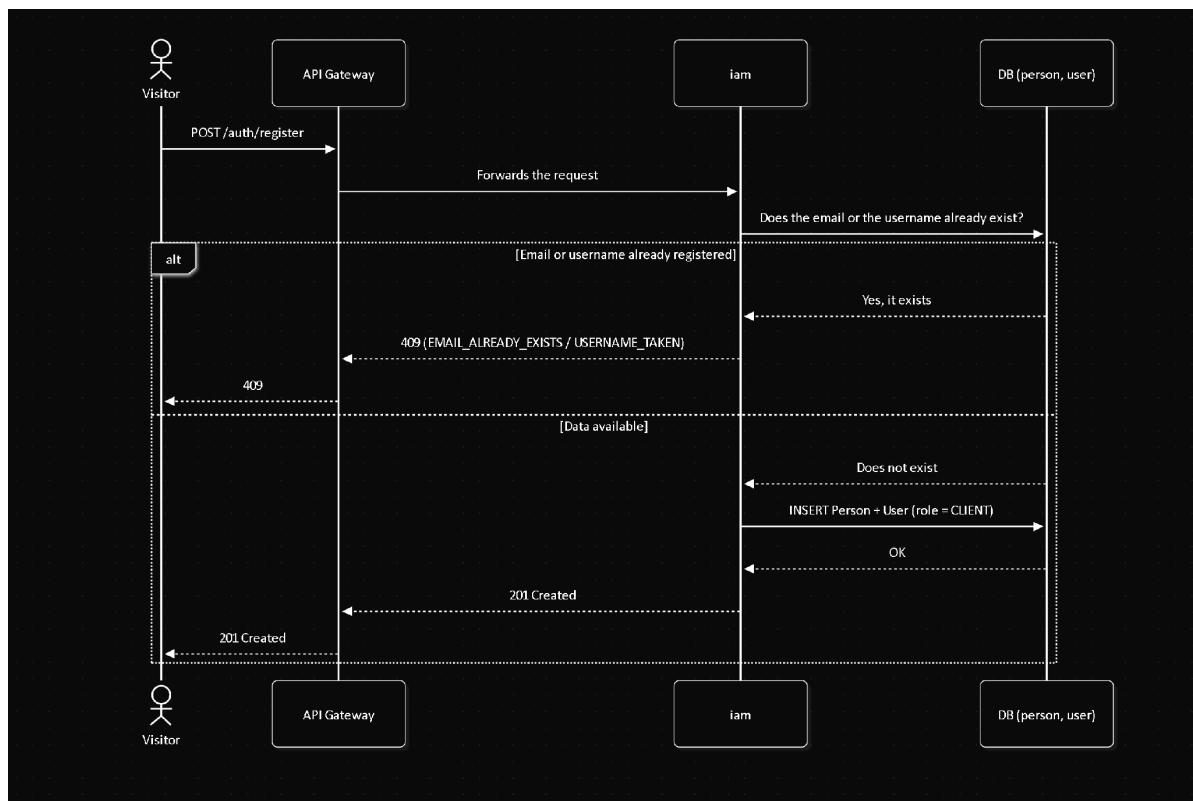
1. Introduction

This document presents the sequence diagrams of RentaMovil's main flows: the messages exchanged between the actor, the API Gateway, the microservices and the database for each operation. It is the second of the four dynamic views of the software analysis; the use cases, the activity diagrams and the state diagrams are in separate documents inside this same folder.

Note on service names: the diagrams use the physical deployment names of the microservices (api-gateway, iam, fleet-maintenance, booking-reservation, payment-billing, telemetry-gps), not the conceptual Bounded Context names from the SRS. The Rental Execution and Notification flows, which appear as their own contexts in the SRS, physically run inside booking-reservation — they are part of the same lifecycle and don't need to be deployed as separate services.

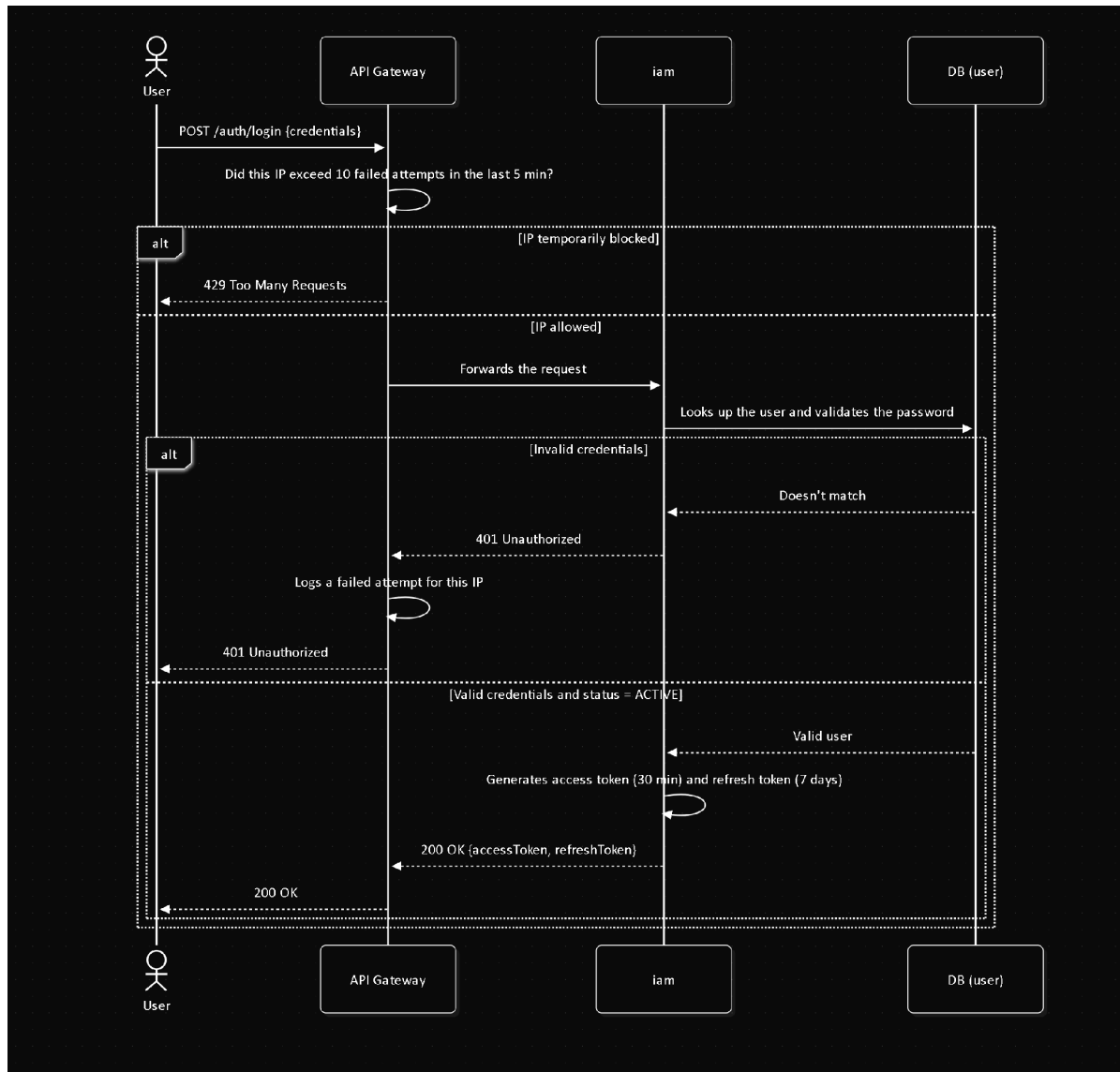
SEQ-01 — Customer registration

Field	Value
Actor	Visitor
Related FR	FR1.1
Precondition	The visitor has no account in the system
Postcondition	A Person and a User are created with the CLIENT role



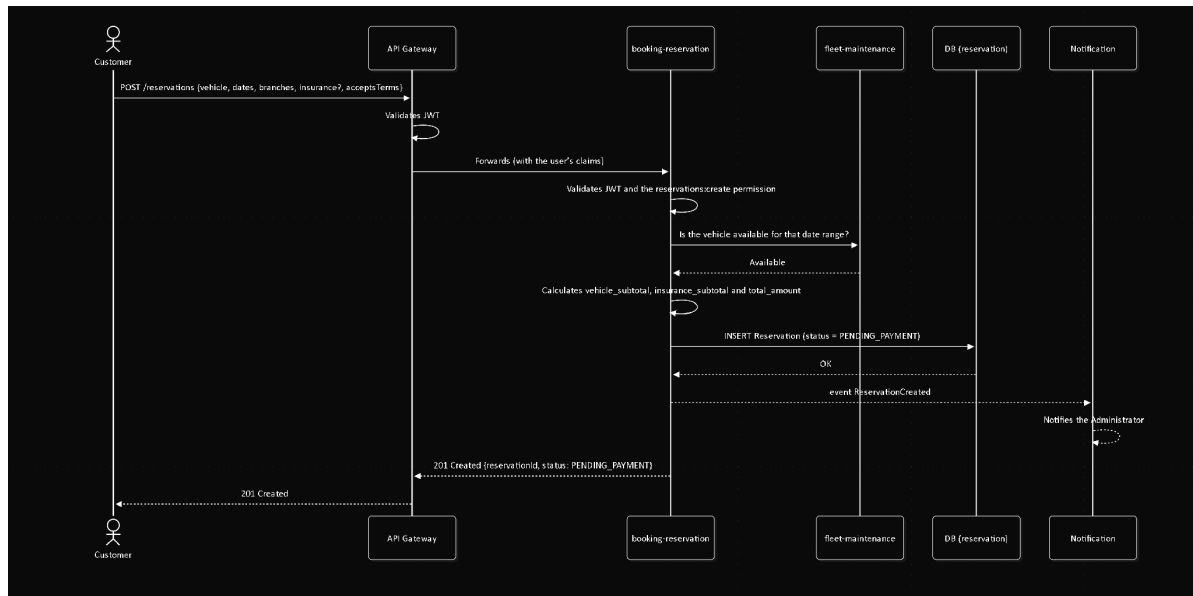
SEQ-02 — Login

Field	Value
Actor	Customer (or any registered user)
Related FR	FR1.2
Precondition	The user has an active account
Postcondition	An access token and a refresh token are issued



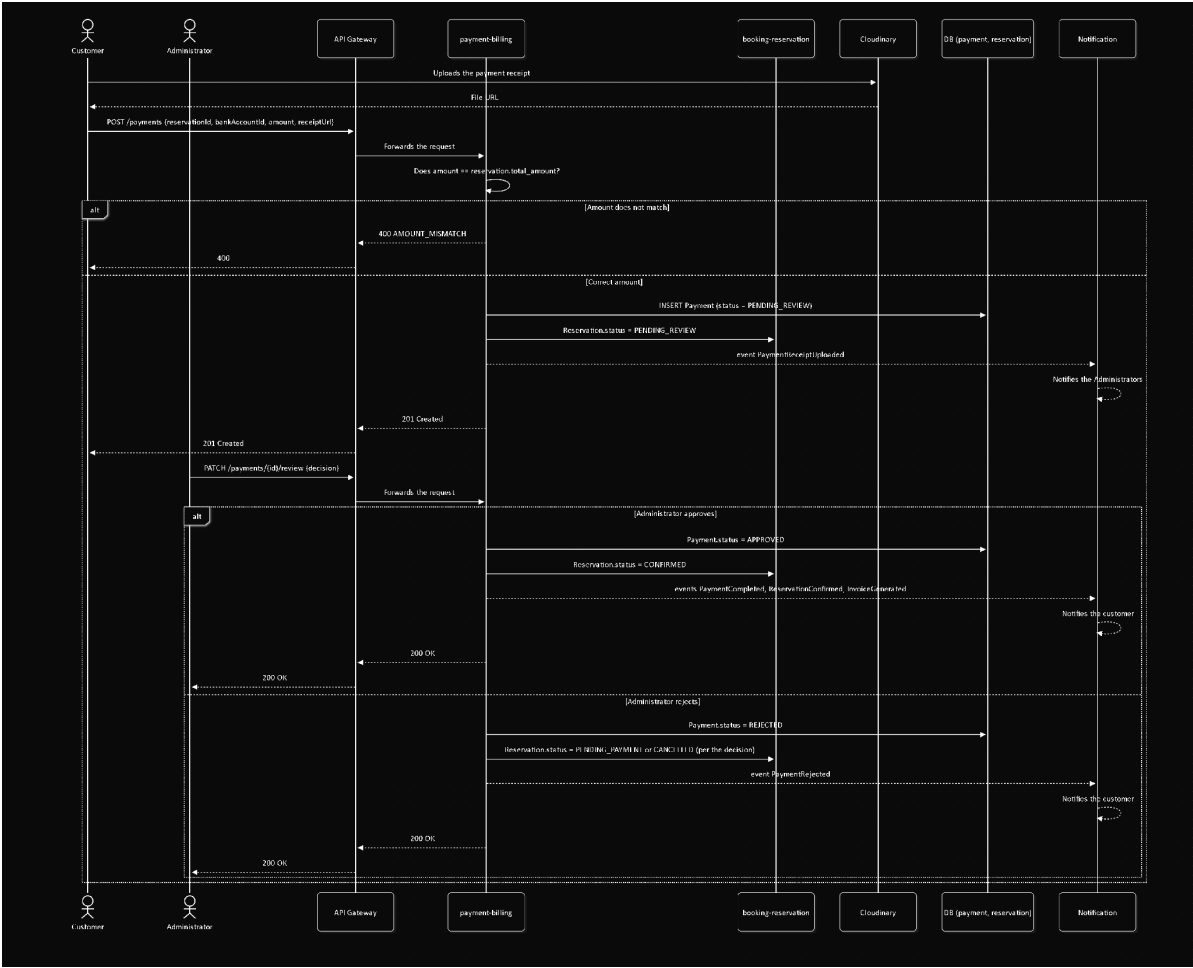
SEQ-03 — Create a reservation

Field	Value
Actor	Customer
Related FR	FR3.1, FR2.2
Precondition	Customer authenticated; the vehicle exists
Postcondition	A Reservation is created in PENDING_PAYMENT ; the Administrator is notified



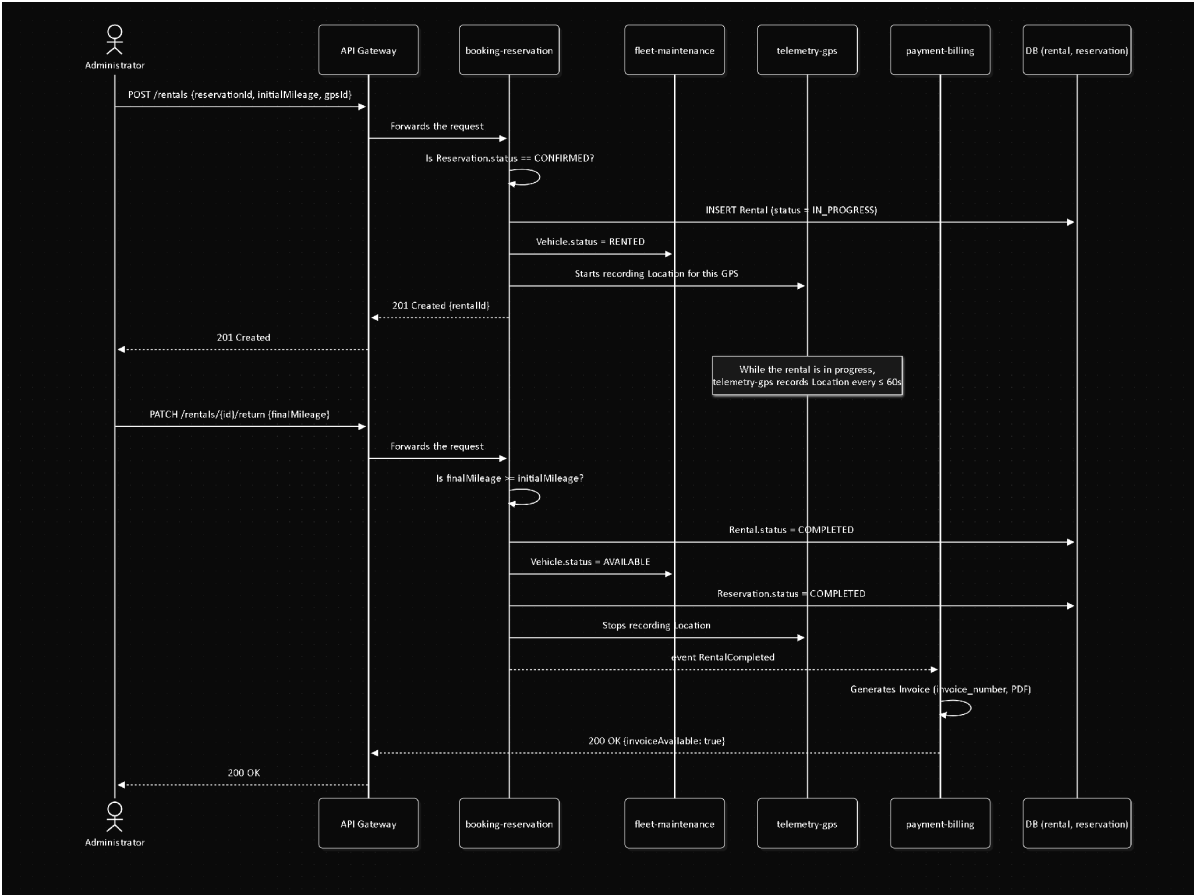
SEQ-04 — Pay for a reservation (receipt review)

Field	Value
Actors	Customer, Administrator
Related FR	FR5.1
Precondition	Reservation in PENDING_PAYMENT
Postcondition	Reservation CONFIRMED (approved) or PENDING_PAYMENT/CANCELLED (rejected)



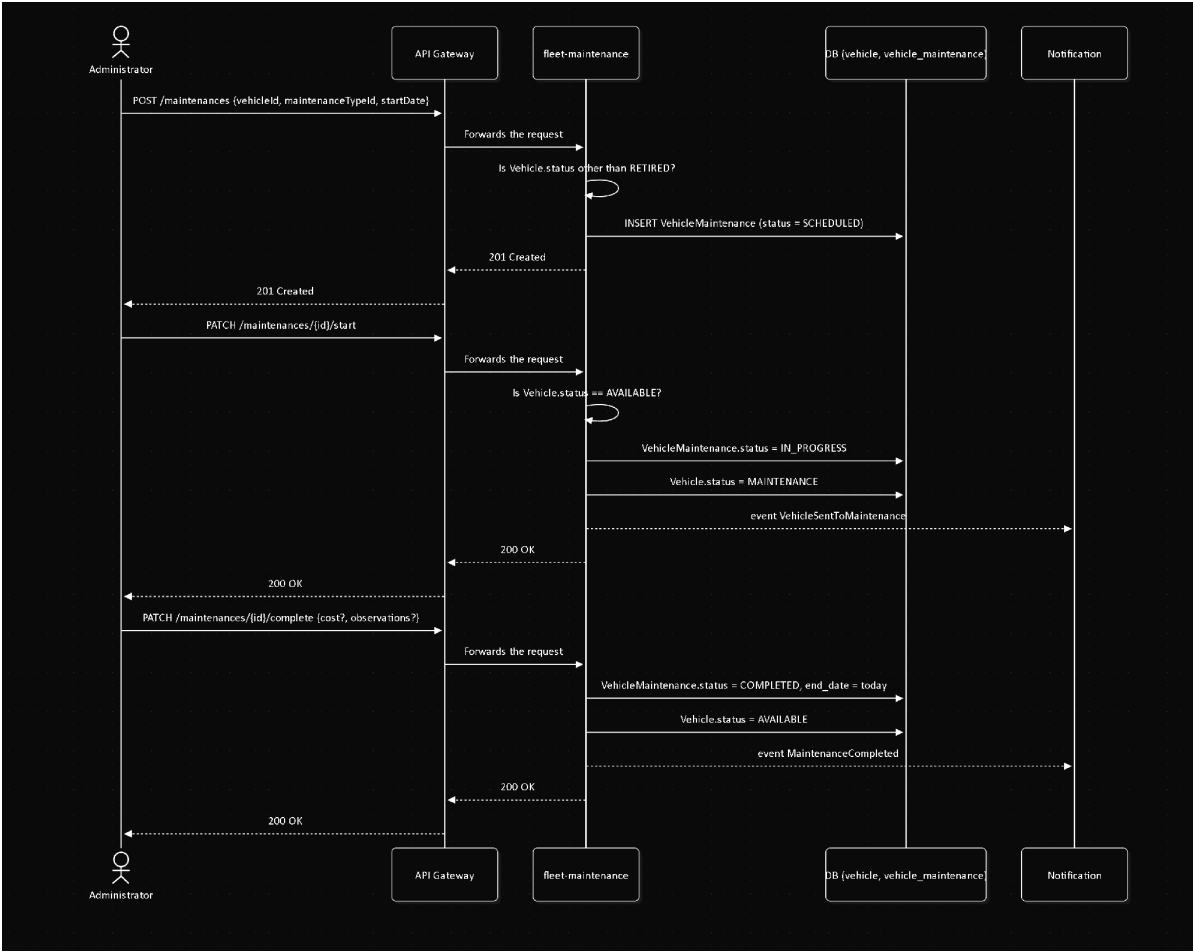
SEQ-05 — Execute the rental (pickup, return and invoice)

Field	Value
Actor	Administrator
Related FR	FR4.1, FR4.2, FR5.2
Precondition	Reservation in CONFIRMED
Postcondition	Rental completed, Vehicle available again, invoice generated



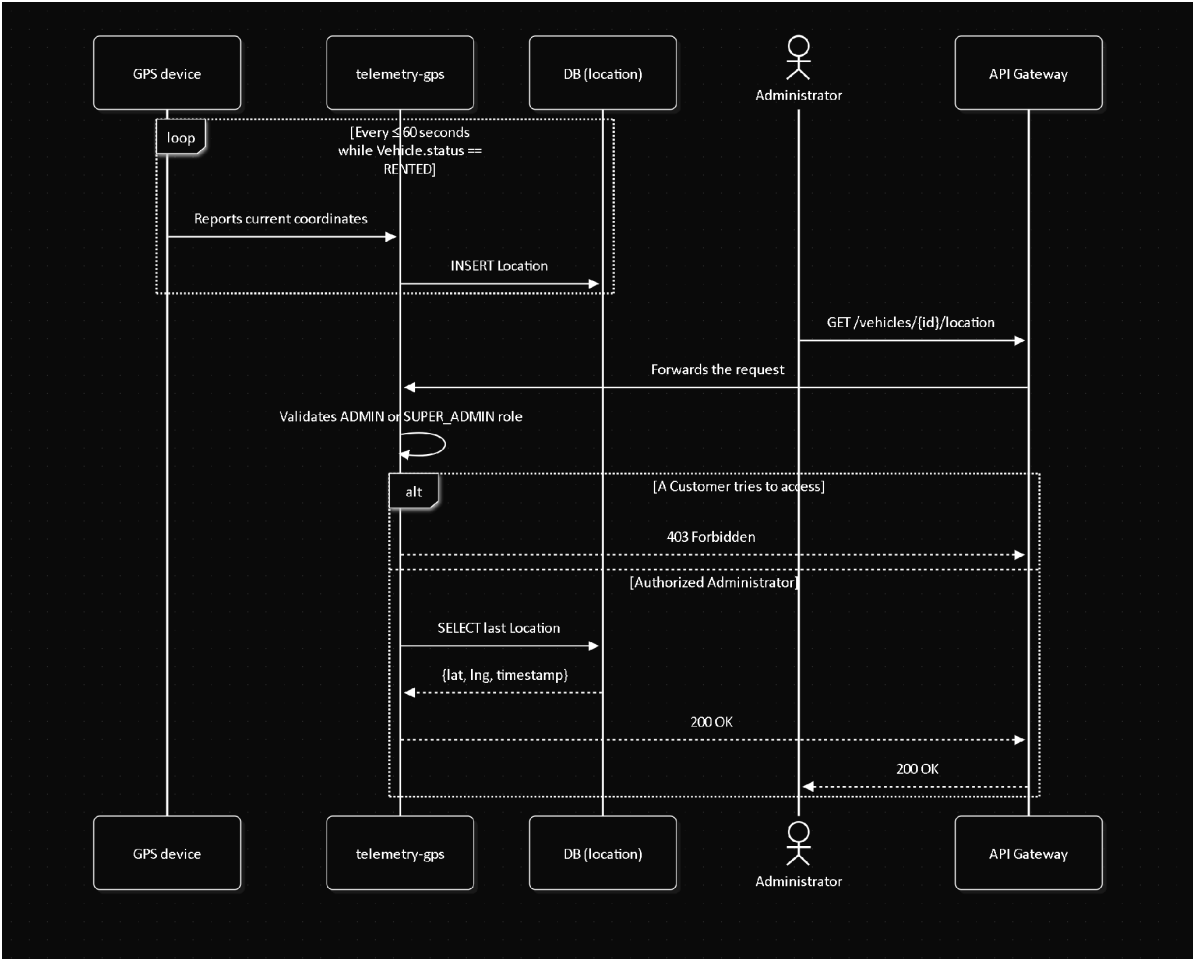
SEQ-06 — Schedule and complete a maintenance

Field	Value
Actor	Administrator
Related FR	FR8.3, FR8.7
Precondition	The vehicle is not RETIRED
Postcondition	Maintenance COMPLETED ; vehicle AVAILABLE again



SEQ-07 — Real-time GPS tracking

Field	Value
Actor	Administrator
Related FR	FR6.1
Precondition	The vehicle is in RENTED status
Postcondition	The last recorded location is returned



References

Document title	Reference
Software Requirements Specification	01-informe-especificacion-requisitos/SR S.md (this repository)
Use cases	01-Use-Cases.md (this folder)
Activity diagrams	03-Activity-Diagrams.md (this folder)
State diagrams	04-State-Diagrams.md (this folder)
Domain events catalog	rtm-docs repository, 02-domain/domain-events.md
Microservices catalog (physical deployment)	rtm-docs repository, 09-microservices/service-catalog.md